2.3.6 Wrap-Up: Error Analysis

1. Errors were made in finding expressions equivalent for $4^{(3s-1)}$. Identify the errors, then write two correct equivalent expressions.

$$4^{(3s-1)} = \frac{4^{(3s)}}{4^1} = \frac{(4^3)^s}{4^1} = \frac{64 \cdot 4^s}{4} = 4 \cdot 4^s$$

Unit 2, Lesson 4: Introducing Polynomials

Benchmark

MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.

Learning Targets

- ☐ I can write polynomials in standard form.
- ☐ I can multiply a monomial to a polynomial.

2.4.1 Warm-Up: Number Puzzle

1. Solve the puzzle by filling in each blank cell in the blue box so that each row and column has the sum shown. Each cell must contain a different value.

		100
		120
105	115	

2.4.2 Exploration: Defining a Polynomial

1. The charts show several examples and non-examples of polynomials.

Polynomials

$r^5 + 2.7r^3 - 0.3r$

$\frac{3}{4}x^2y$ $\quad 80z^{15}$

$b - \frac{1}{5}$ $\quad 58c^0$ $\quad \frac{24k}{3}$

$\sqrt{9}np$ $\quad (m - 16)$

Not Polynomials

$r^5 + 2.7r^{-3} - 0.3r$

$\frac{3x^2}{4y}$ $\quad 80z^{1.5}$

$\frac{1}{b-5}$ $\quad 58c^x$ $\quad 24k^{\frac{1}{2}}$

$\sqrt{9np}$ $\quad (m - 16)^{-1}$

a. Create a list of identifying features polynomials can have versus those polynomials cannot have.

Polynomials can have …	Polynomials cannot have …

b. Compare your lists with your partner. Add any additional features you found after your discussion.

c. With your partner, write a definition for polynomial.

d. Ask two other pairs of partners the definition they wrote. Listen to their definition and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary of their definition was accurate.

My Summary of the Definition of a Polynomial	**Initials**

e. After reviewing the definitions created by your classmates, go back to your definition and make revisions, if needed.

Some polynomials are described with specific names based on the number of terms.

2. Four prefix options are given.

Prefixes			
tri-	mono-	poly-	bi-

a. Match the prefixes to identify the types of polynomials in the table based on the description and examples.

Polynomials		
_______mials (one term)	**______nomials** (two terms)	**_______nomials** (three terms)
$-7.8t^3w$	$-7.8t^3w+2.04w$	$-7.8t^3w+2.04w-t$
14	$-\frac{8}{9}y^2+14$	$-\frac{8}{9}y^2-12m+14$

b. In the last row of the table, create your own polynomial that fits each category.

2.4.3 Guided Instruction: Polynomials and Standard Form

A ***monomial*** is a polynomial with one term.

The **degree of a monomial** is the sum of the exponents of the variables in a monomial.

1. Identify the degree of each monomial.

Monomial	Degree
$-4x^5y^6$	
$1.6p^6c^5w^8$	
$-\frac{7}{11}rh^{24}$	

A ***polynomial*** is the sum or difference of terms which have variables raised to non-negative integer powers and which have coefficients that may be real or complex.

The **degree of a polynomial** is the greatest degree of the monomials in a polynomial.

2. What is the degree of $62pk^8+7a^2b^5-38g^4h^6$?

The **leading coefficient** is the coefficient of the first term of a polynomial whose terms are written in descending order from largest degree to smallest degree.

3. What is the leading coefficient of $62pk^8 + 7a^2b^5 - 38g^4h^6$?

A polynomial in ***standard form*** is one where the term with the highest degree is written first and the other terms are in descending order by degree. The variables in each term should be written in alphabetical order.

4. Rewrite polynomial $62pk^8 + 7a^2b^5 - 38g^4h^6$ in standard form.

2.4.4 Collaboration: Polynomials and Standard Form

Work with your partner to complete the following.

1. Three polynomials are given in the table.

   a. Determine whether each polynomial is written in standard form. If it is not, rewrite it in standard form.

   b. Then, in the second column, identify the features of each polynomial.

$-52a + 3a^3b^4 + 21a^2b^2$	Number of terms _______ Degree of the polynomial _______ Leading term _________ Leading coefficient _________
$-9x^8y^4 + 2x^9y^2$	Number of terms _______ Degree of the polynomial _______ Leading term _________ Leading coefficient _________
$5m^6 + 13m^2 - \frac{1}{4}m^8$	Number of terms _______ Degree of the polynomial _______ Leading term _________ Leading coefficient _________

2. With your partner, match the polynomial in the left column by drawing an arrow to its descriptive feature in the right column.

$x^3 + 4x^2 - 5x + 9$	fifth-degree polynomial
$45a^2b^3$	constant term of -2
$3x^4 - 9x^3 + 4x^9$	seventh-degree polynomial
$7a^6b^2 + 18ab^3c - 9a^7$	leading coefficient of 3
$x^5 - 9x^3 + 2x^7$	four terms
$3x^3 + 7x^2 - 11$	eighth-degree polynomial
$x^2 - 2$	equivalent to $4x^9 + 3x^4 - 9x^3$

2.4.5 Refresher: Distributing a Monomial to a Polynomial

The ***distributive property of multiplication over addition*** applies to polynomials. Recall that the distributive property "distributes" a factor to all terms within a set of parentheses.

1. An example of distributing a monomial to a polynomial is shown.

$$-3b^2\left(\frac{1}{4}m^3 + 2m - 8\right)$$

$$-\frac{3}{4}b^2m^3 - 6b^2m + 24b^2$$

   a. Discuss with your partner what operation the arrows are demonstrating in the example.

   b. In the resulting product, why are the first two terms negative while the third term is positive?

2.4.6 Guided Practice: Distributing a Monomial to a Polynomial

1. Find each product by distributing the monomial to the polynomial. Then, complete the table for the resulting product.

$(1+3y^3-2y^4)(-6y^4)$	
Product in Standard Form	
Leading Coefficient	Degree of Polynomial

$-6dg^4(d^9g^2-2d^4g^2+10)$	
Product in Standard Form	
Leading Coefficient	Degree of Polynomial

$\frac{3}{2}a^2\left(8a-9b^2+\frac{1}{3}a^3b\right)$	
Product in Standard Form	
Leading Coefficient	Degree of Polynomial

$-0.25b^3c(1.5c+6b^2-4b^5c^3)$	
Product in Standard Form	
Leading Coefficient	Degree of Polynomial

2.4.7 Wrap-Up: Growth Mindset

1. Mistakes are an opportunity to learn. Complete the statement to highlight a mistake you learned from.

   My favorite mistake from today was …

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Unit 2, Lesson 5: Adding and Subtracting Polynomials

Benchmark

MA.912.AR.1.3 Add, subtract and multiply polynomial expressions with rational number coefficients.

Learning Target

- [ ] I can add and subtract polynomials.